AI Opportunity Report: Cybit

AI/ML Engineer Intern task: Identify an AI Opportunity for Cybit

Scope note. The brief names "Cynbit". I could not find a company by that name, so I assumed it means **Cybit** (cybit.com, Tyneside, UK). All company facts come from public web profiles; no internal data was available. Anything about Cybit's internal volumes or costs is an **assumption** and is labelled as such.

1. Executive summary

Cybit is a specialist technology solutions provider that sells data and AI, modern workspace, hybrid cloud and security services, largely to small and medium businesses. Like any managed service provider (MSP), its main cost is skilled engineers' time spent on service-desk tickets and security alerts.

Recommendation: build an **AI ticket and alert triage copilot with runbook retrieval**. It classifies incoming tickets, suggests a priority, finds the right runbook, drafts an engineer summary and a customer reply, and sends uncertain or risky cases to a human. It is the best mix of business value, buildability and low risk among the options I assessed, and I built a working proof of concept (see the Technical Approach Report).

2. Company snapshot (public information)

- Positions itself as a specialist technology solutions provider covering **data and AI, modern workspace, hybrid cloud and security**.
- Public profiles list managed IT services, a network operations centre, cybersecurity, Azure and AWS, Check Point, Qlik, Alteryx, Microsoft Dynamics and business intelligence.
- Recent public video content shows a push into **continuous security validation**, including a partner webinar and a MITRE ATT&CK threat-simulation video, alongside customer stories.
- Headcount estimates differ widely between data aggregators (from under 10 to roughly 200), so I do not rely on any size or revenue figure.

3. Industry context

MSPs and managed security providers sell human attention at scale: monitoring, service desk, security operations and projects. Three pressures make AI relevant:

1. **Volume grows faster than headcount.** Every new customer adds endpoints, cloud tenants and security tools, and every tool generates alerts.
2. **Skilled engineers are scarce and expensive.** Senior staff spend time on repetitive first-line work: categorising a ticket, finding the runbook, writing the first update.
3. **Customers expect speed and plain language,** especially SMEs without their own security team.

Sector-specific risks: customer data confidentiality, contractual SLAs, and the cost of a wrong action during a security incident. AI here must be **assistive, auditable and human-approved**, not autonomous.

4. Opportunity long-list

Scores are my own judgement on a 1-5 scale (5 = best). Feasibility means buildable and testable by a small team in weeks using data Cybit already holds.

#	Opportunity	Business value	Feasibility	Data risk	Verdict
1	**Ticket and alert triage with runbook copilot** (service desk, NOC, SOC)	5	5	Medium	**Chosen**
2	Plain-language security validation and remediation reports	4	4	Medium	Strong follow-on; depends on partner tool outputs
3	Customer-facing service desk assistant	4	3	High	Later phase, after internal copilot builds trust
4	Churn and upsell prediction from CRM and ticket history	3	2	Medium	Needs clean historical data; poor fit for a take-home
5	Marketing repurposing from webinar and video transcripts	2	5	Low	Easy but low strategic impact

Why not the others (yet)

- **Security validation reports (#2)** are attractive given Cybit's public focus on attack simulation, but their value depends on the output format of third-party tools I cannot see.
- **A customer-facing assistant (#3)** puts model errors directly in front of customers. It is much safer to prove the same engine internally first.
- **Churn prediction (#4)** needs real, labelled customer history, which a prototype cannot honestly fake.
- **Marketing repurposing (#5)** is safe but does not touch the core cost base.

5. Chosen opportunity: ticket triage and runbook copilot

The problem

Each incoming ticket or alert is read by a person who must decide *what is this, how urgent is it, who owns it, what do we do first, and what do we tell the customer*. This is repetitive, SLA-sensitive, and inconsistent between engineers and shifts.

The solution

1. **Classify** the ticket into a category (account access, phishing, endpoint alert, network, cloud, backup, Microsoft 365, hardware, data and BI, access change).
2. **Prioritise** it (P1-P4) with explicit severity signals such as scope, active attack and business impact.
3. **Retrieve** the most relevant runbooks.
4. **Draft** an engineer summary, numbered next steps and a customer reply, grounded only in the retrieved runbooks.
5. **Gate:** always route P1s, low-confidence cases and "low priority but alarming wording" cases to a human.

Illustrative value model (assumptions, not Cybit data)

Assumption	Value
Tickets and alerts handled per month	3,000
Minutes of triage and first-response work per ticket today	6
Share of tickets where the draft is accepted with minor edits	50%
Minutes saved per accepted ticket	4

Result: about 3,000 x 50% x 4 min = **100 engineer-hours per month**, plus more consistent prioritisation and faster first response. The real figure must be measured in a pilot; the point of this model is to show which inputs matter.

Success metrics for a pilot

- Category and priority accuracy against engineer-confirmed labels
- **P1 recall** (the share of true P1s that are escalated; the most important safety metric)
- Share of tickets handled without human correction

- Median time to first response and SLA breach rate
- Engineer acceptance rate of drafted replies and runbook suggestions

6. Risks and mitigations

Risk	Mitigation
Customer data leaves Cybit's control	Use a private or contractually protected model endpoint; redact personal data before the model call
Wrong priority delays a real incident	Human review for all P1s and low-confidence cases; keyword escalation; monitor P1 recall
Over-escalation causes alert fatigue	Track false P1 rate; tune thresholds per customer
Model invents steps or commands	Ground answers in retrieved runbooks only; cite runbook IDs; refuse when nothing matches
Prompt injection via ticket text	Treat ticket text as data, never as instructions; no tool access in the first release
Drift as customers and tools change	Weekly accuracy review; retrain from engineer corrections

7. Roadmap

1. **Weeks 1-2: Shadow mode.** Run on historical tickets; compare with engineer decisions; no customer impact.
2. **Weeks 3-6: Assist mode.** Engineers see suggestions in the ticketing tool and accept or edit them; capture corrections as new labels.
3. **Weeks 7-12: Extend.** Add alert sources from the NOC and SOC, then use the same retrieval engine for security validation reports.
4. **Later:** consider a customer-facing assistant with strict scope and escalation.

8. Conclusion

Triage is where an MSP's cost, SLA risk and customer experience meet. A grounded, human-approved copilot is a low-risk first AI product that also builds the data and trust Cybit would need for its other AI ideas.